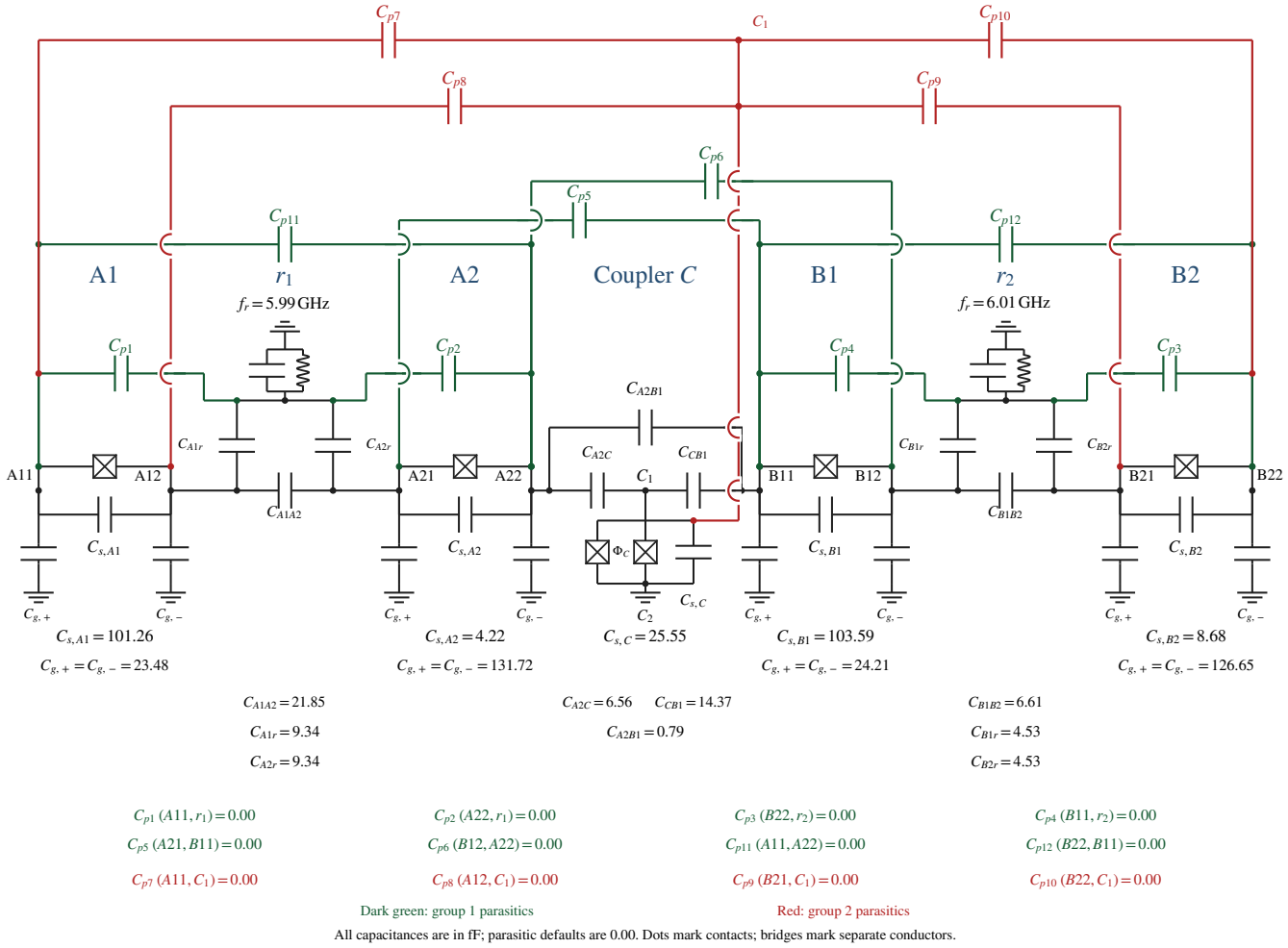
Parasitic compensation | validated results

Grounded coupler with floating rail transmons and both readout resonators. The report uses the selected 1-ppt minimum-change geometries for all 24 cases, each containing one 0.1 or 0.2 fF parasitic. Every gate and OFF result below was recalculated for that same capacitor allocation.



The circuit shows the nominal physical capacitor values. Green Cp1-Cp6/Cp11-Cp12 form group 1, and red Cp7-Cp10 form group 2. Actual compensations are shown later in percent and fF. The unrounded nominal input, fitted circuits and complete validation data are embedded.

Input or comparison control	Value
Loaded rail EC/h	A1,A2,B1,B2 = 165,250,160,260 MHz
Loaded coupler EC/h	450.285 MHz
Nominal audit-basis OFF	9.89676601 GHz
Static gate controls	Coupler 7.05 GHz, duration 100.3 ns. Common B frequency corrects conditional phase to pi.
Computed validation outcomes	24/24 OFF points below 10 GHz meet the 1-Hz within-basis threshold. 24/24 gate calibrations meet phase pi.

Minimum-change compensation protocol

Inputs and circuit reduction. Each of the 24 independent cases starts from the unrounded nominal physical record embedded as nominal-compensation-input.json. Add one registered parasitic at 0.1 or 0.2 fF. A capacitor c between nodes a,b contributes +c to Caa,Cbb and -c to Cab,Cba. Ground C2 and build the 11-node matrix. The charge embedding P places +1/-1 on floating transmon pad pairs, +1 on C1 and +1 on each resonator node. Common-mode charge of each floating transmon is fixed to zero. Compute $K = P^T C_{node}^{-1} P$ in order A1,A2,C,B1,B2,r1,r2. In SI units, $ECi/h = e^2 Kii/(2h)$.

Adjustable geometry. Write Ck = Ck,0(1+xk). The 20 fractional variables comprise four rail ground pairs, nine intentional links, two resonator self capacitors and five transmon across-pad self capacitors. Both ground capacitors of each floating transmon move together. The added parasitic and resonator inductances stay fixed. All seven K diagonals equal their nominal values, preserving all five loaded ECs and the two bare resonator frequencies. Positive physical capacitors and a numerical search guard |xk| <= 0.95 apply.

Matching tolerance. Set epsilon = 0.001, or 1 ppt = 0.1% = 1000 ppm. For each intentional pair require |Kij-Kij,0|/|Kij,0| <= epsilon. The nine pairs are A1-A2, A2-C, C-B1, A2-B1, B1-B2, A1-r1, A2-r1, B1-r2 and B2-r2. Also require $||K-K0||_{F,M}/||K0||_{F,M}$ <= epsilon. Group 1 includes every entry in the symmetric mask M. Group 2 includes the seven diagonals and nine intentional pairs. Its twelve other off-diagonal entries remain residual diagnostics. Diagonal constraints are exact numerical equalities. Near-zero and excluded channels require absolute-coupling and quantum checks.

Stage 1. Minimize the worst capacitor change. Introduce t with -t <= xk <= t and 0 <= t <= 0.95. Minimize $t + 10^{-8}\ \mathrm{sum}(xk^2)$ subject to the diagonal, individual-link and matched-matrix constraints above. The engineering matrix tolerance is a constraint. This explicitly favors smaller geometry changes once adequate matching is achieved.

Stage 2. Reduce combined movement. With the best t = t*, set the ceiling to $t^* + \max(10^{-10}, 10^{-6} t^*)$, clipped to 0.95. Minimize $\mathrm{sum}(xk^2)$ within that ceiling and all matching constraints. This resolves equivalent minimax allocations toward smaller total movement. Local convergence is checked for both stages. The results establish feasible local fits.

Numerical reproduction. Use studies.bounded_parasitic_compensation.compensate_bounded(physical, ports, amount, max_fraction=0.95, match_rtol=0.001, excluded_mode_pairs=excluded). Use excluded=() for group 1. For group 2 exclude the twelve pairs outside its intentional set. Start from an EC-preserving self-only adjustment. SLSQP uses analytic Jacobians, ftol=1e-11/maxiter=500 in stage 1 and ftol=1e-12/maxiter=200 in stage 2. For incidence vector b, $dK/dc = -P^T C^{-1} b b^T C^{-1} P$. Verify EC error below 1 Hz, fixed diagonals, positive capacitors and both matching inequalities after rebuilding the physical network. The exact solver and audit JSON are embedded.

Meaning of the tolerance and result status. The 1-ppt value is a configurable engineering fit target. EM and fabrication uncertainty require separate estimates. All 24 new electrostatic fits and secondary stages converged. All 24 quantum validations have been computed for these geometries. OFF and phase-pi acceptance counts are on page 1. Basis comparisons and unresolved numerical qualifications are reported explicitly on the final page.

Gate and OFF results | 0.1 fF

Geometry and idle rail EJs stay fixed. Each OFF point tunes coupler EJ alone and targets physical 0100-0010 exchange. Static CZ uses consistently dressed ON logical references, fixed coupler 7.05 GHz and 100.3 ns, with common-B phase correction. Raw and postselected scores use the same phase-calibrated operating point.

Parasitic	Max cap change	Matched K error (ppm)	OFF f01 (GHz)	Raw CZ	Postselected CZ
A21-B11	10.2%	568.5	9.93661671	99.363%	99.9942%
A11-r1	1.4%	118.0	9.88280317	99.373%	99.9951%
A22-r1	0.7%	93.7	9.88281452	99.362%	99.9953%
B22-r2	2.1%	146.9	9.91085538	99.438%	99.9950%
B11-r2	2.1%	156.6	9.88263237	99.378%	99.9946%
B12-A22	12.6%	687.1	9.93686443	99.427%	99.9954%
A11-A22	1.3%	121.3	9.91081184	99.427%	99.9947%
B22-B11	2.0%	387.8	9.91122753	99.380%	99.9954%
A11-C1	0.4%	75.6	9.87629223	99.512%	99.9949%
A12-C1	0.4%	75.6	9.87446252	99.142%	99.9942%
B21-C1	0.4%	75.7	9.88229597	99.406%	99.9938%
B22-C1	0.5%	75.7	9.91136754	99.352%	99.9957%

Nominal raw / postselected scores are 99.399% / 99.9949%. Postselected score equals raw fidelity divided by mean logical survival. Gate propagation is the coherent static-ON benchmark. Full scores, survival, per-input leakage, frequency trims and unrounded controls are embedded.

All rates use the recorded numerical bases. Differences below 0.01 percentage point in gate fidelity carry the agreed basis precision qualification. OFF frequencies are explicitly evaluated settings. The final page separates within-basis cancellation from cross-basis convergence.

Gate and OFF results | 0.2 fF

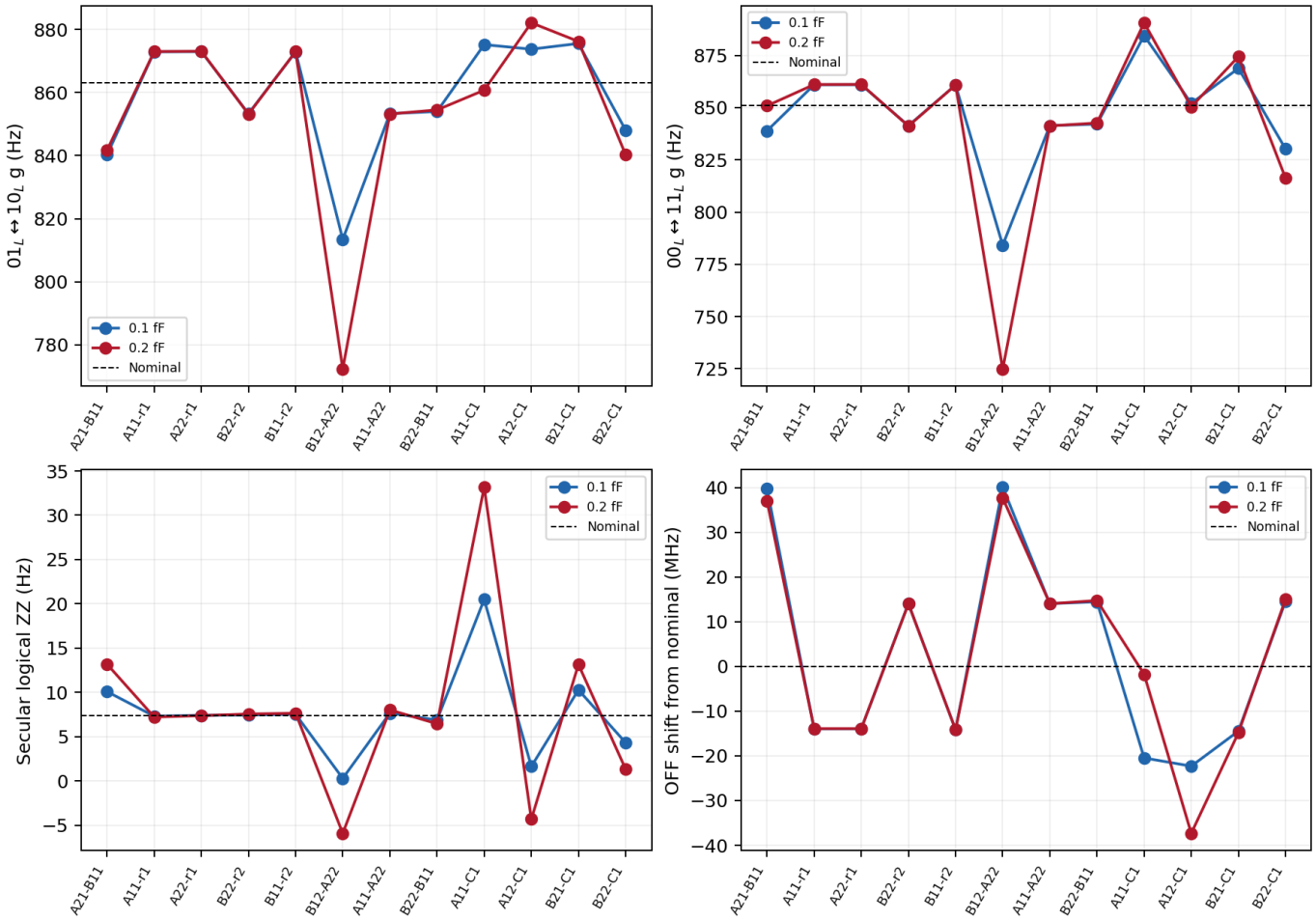
Geometry and idle rail EJs stay fixed. Each OFF point tunes coupler EJ alone and targets physical 0100-0010 exchange. Static CZ uses consistently dressed ON logical references, fixed coupler 7.05 GHz and 100.3 ns, with common-B phase correction. Raw and postselected scores use the same phase-calibrated operating point.

Parasitic	Max cap change	Matched K error (ppm)	OFF f01 (GHz)	Raw CZ	Postselected CZ
A21-B11	21.4%	1000.0	9.93379296	99.268%	99.9935%
A11-r1	2.8%	188.4	9.88285590	99.364%	99.9952%
A22-r1	1.6%	129.7	9.88286045	99.357%	99.9954%
B22-r2	4.3%	274.3	9.91076294	99.448%	99.9949%
B11-r2	4.3%	294.8	9.88260597	99.388%	99.9945%
B12-A22	26.4%	1000.0	9.93443170	99.485%	99.9962%
A11-A22	2.7%	199.5	9.91079884	99.442%	99.9947%
B22-B11	4.0%	759.0	9.91150814	99.320%	99.9955%
A11-C1	0.9%	52.1	9.89502224	99.167%	99.9921%
A12-C1	0.8%	75.6	9.85943112	98.772%	99.9912%
B21-C1	1.0%	75.7	9.88192720	99.386%	99.9925%
B22-C1	1.1%	75.7	9.91179131	99.282%	99.9962%

Nominal raw / postselected scores are 99.399% / 99.9949%. Postselected score equals raw fidelity divided by mean logical survival. Gate propagation is the coherent static-ON benchmark. Full scores, survival, per-input leakage, frequency trims and unrounded controls are embedded.

All rates use the recorded numerical bases. Differences below 0.01 percentage point in gate fidelity carry the agreed basis precision qualification. OFF frequencies are explicitly evaluated settings. The final page separates within-basis cancellation from cross-basis convergence.

Logical interactions at individual OFF points



All logical interactions are evaluated at the individually retuned physical OFF point, with saved idle rail EJs. Logical exchanges use the canonical four-logical-state projection followed by local single-rail energy-axis alignment. Secular ZZ = (E11-E10-E01+E00)/h uses tracked dressed eigenenergies. Virtual auxiliary, coupler and resonator effects are retained.

OFF shift is fC,OFF(case) - 9.89676601 GHz. Positive means a higher required coupler frequency. The frequencies are verified by direct exchange evaluation. Logical exchange and ZZ remain residual diagnostics alongside the physical cancellation target.

Idle residuals and detunings | 0.1 fF

Parasitic	Physical g at nominal OFF (kHz)	01L-10L g (Hz)	|detuning| (MHz)	00L-11L g (Hz)	|detuning| (MHz)	Secular ZZ (Hz)
A21-B11	54.330	840.3	53.175	838.9	239.236	10.1
A11-r1	-19.230	872.9	52.616	860.9	239.177	7.3
A22-r1	-19.215	873.0	52.885	861.0	239.506	7.4
B22-r2	19.283	853.3	52.563	841.4	239.831	7.4
B11-r2	-19.468	872.9	52.456	860.8	239.331	7.5
B12-A22	54.663	813.4	53.277	784.2	239.127	0.3
A11-A22	19.222	853.3	52.566	841.4	239.774	7.7
B22-B11	19.791	854.0	53.027	842.1	239.367	6.9
A11-C1	-28.190	875.2	52.281	884.5	239.389	20.5
A12-C1	-30.717	873.8	52.798	852.0	239.908	1.6
B21-C1	-19.927	875.6	51.866	868.8	239.925	10.3
B22-C1	19.984	848.1	53.391	830.3	239.000	4.3

The physical-exchange column keeps the common nominal audit-basis OFF frequency. The logical-exchange and ZZ columns use each candidate's individually retuned OFF frequency. Every idle rail EJ remains fixed. These two bias conventions are kept separate to show both the effect before retuning and the recovered result.

Parasitic	Retuned physical g (Hz)	Mean logical survival (%)	Worst gate nonlogical population (%)	B phase trim (MHz)
A21-B11	-0.000	99.368	1.554	+0.269
A11-r1	-0.000	99.378	1.153	-0.116
A22-r1	-0.000	99.366	1.238	+0.027
B22-r2	-0.000	99.443	1.117	-0.076
B11-r2	-0.000	99.383	1.158	+0.016
B12-A22	0.000	99.431	1.316	+0.323
A11-A22	0.000	99.433	1.195	+0.092
B22-B11	-0.000	99.385	1.342	+0.080
A11-C1	-0.000	99.517	0.956	-0.260
A12-C1	-0.001	99.147	1.823	+0.101
B21-C1	-0.000	99.412	1.018	-0.193
B22-C1	-0.000	99.356	1.455	+0.213

Survival, nonlogical population and the B trim refer to the static CZ operating point. Mean logical survival averages over the four logical inputs. The worst nonlogical population is the largest leakage among those inputs. The B trim is relative to the nominal phase-calibrated B setting.

Idle residuals and detunings | 0.2 fF

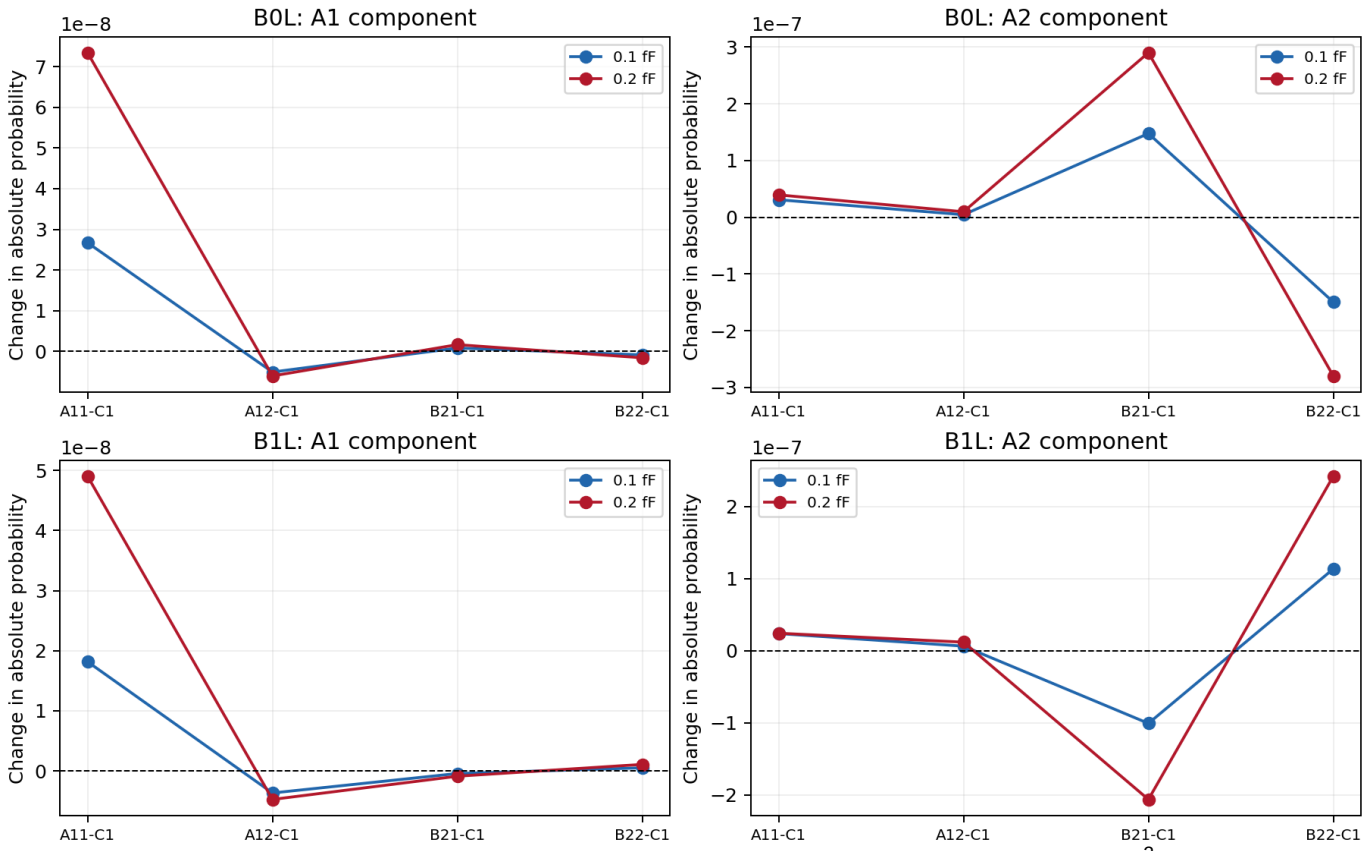
Parasitic	Physical g at nominal OFF (kHz)	01L-10L g (Hz)	|detuning| (MHz)	00L-11L g (Hz)	|detuning| (MHz)	Secular ZZ (Hz)
A21-B11	50.512	841.7	53.561	850.9	238.867	13.2
A11-r1	-19.157	873.0	52.688	861.1	239.106	7.2
A22-r1	-19.153	873.1	52.926	861.1	239.464	7.4
B22-r2	19.157	853.2	52.444	841.2	239.952	7.6
B11-r2	-19.507	873.0	52.368	860.8	239.417	7.7
B12-A22	51.373	772.3	53.557	724.9	238.859	-5.9
A11-A22	19.205	853.2	52.476	841.3	239.831	8.0
B22-B11	20.175	854.5	53.370	842.6	239.026	6.5
A11-C1	-2.393	860.8	51.909	890.6	239.064	33.2
A12-C1	-51.600	882.1	52.928	850.4	240.096	-4.3
B21-C1	-20.434	876.2	51.195	874.4	240.593	13.1
B22-C1	20.566	840.3	54.110	816.3	238.283	1.3

The physical-exchange column keeps the common nominal audit-basis OFF frequency. The logical-exchange and ZZ columns use each candidate's individually retuned OFF frequency. Every idle rail EJ remains fixed. These two bias conventions are kept separate to show both the effect before retuning and the recovered result.

Parasitic	Retuned physical g (Hz)	Mean logical survival (%)	Worst gate nonlogical population (%)	B phase trim (MHz)
A21-B11	0.000	99.275	1.811	+0.403
A11-r1	-0.000	99.369	1.186	-0.091
A22-r1	-0.000	99.361	1.261	+0.040
B22-r2	0.000	99.453	1.072	-0.116
B11-r2	-0.000	99.394	1.120	-0.010
B12-A22	0.000	99.489	1.087	+0.435
A11-A22	0.000	99.448	1.151	+0.055
B22-B11	0.000	99.325	1.559	+0.197
A11-C1	0.000	99.175	2.030	-0.474
A12-C1	-0.003	98.781	3.202	+0.203
B21-C1	-0.000	99.394	1.138	-0.428
B22-C1	-0.000	99.286	1.679	+0.471

Survival, nonlogical population and the B trim refer to the static CZ operating point. Mean logical survival averages over the four logical inputs. The worst nonlogical population is the largest leakage among those inputs. The B trim is relative to the nominal phase-calibrated B setting.

Group 2 | physical idle admixture

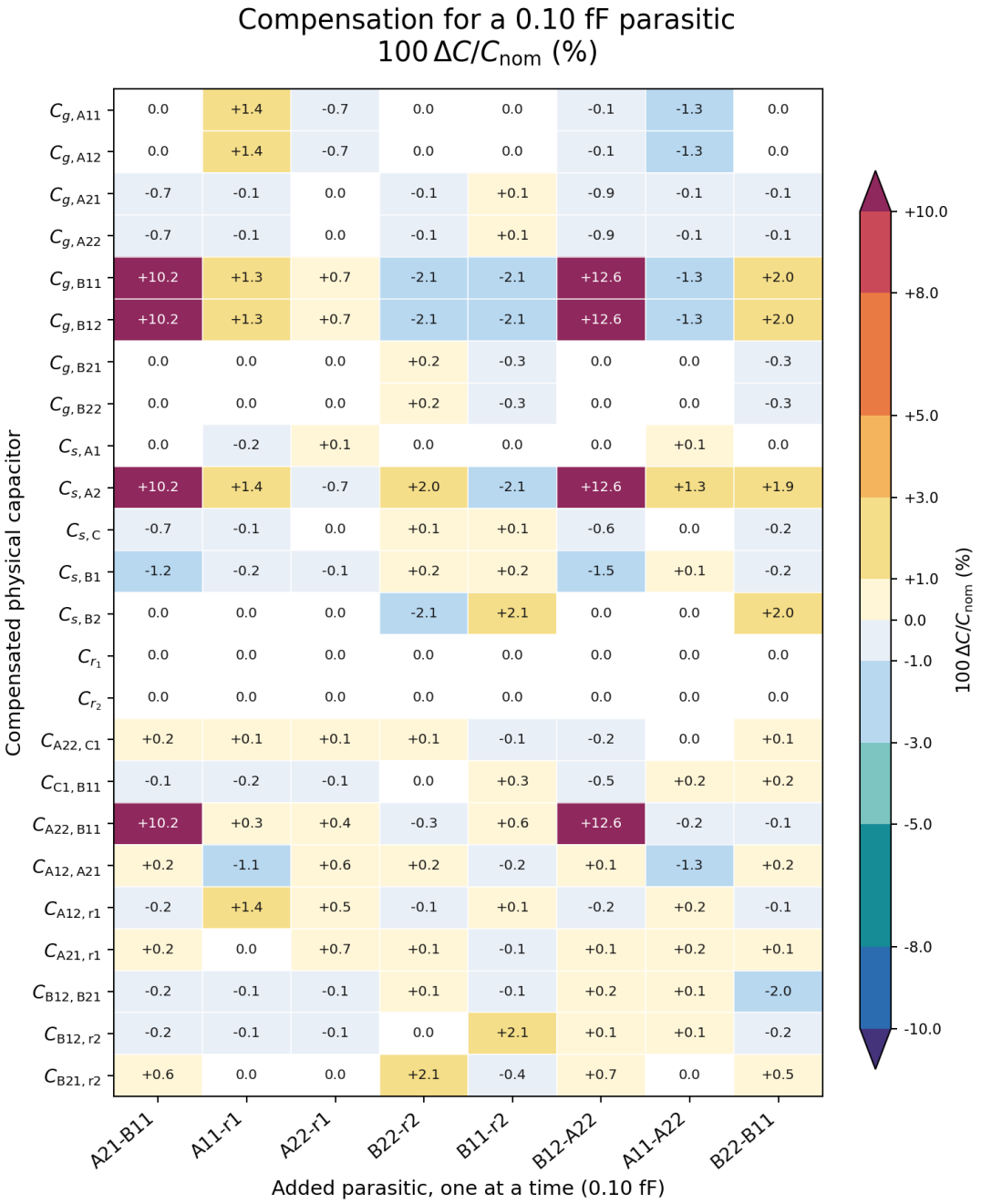


For dressed B0L/B1L, rail A is tracked from its pair-ground reference. A1 weight = |<1000;0C;0r1;0r2|psi>|^2 and A2 weight uses 0100, in rail order A1,A2,B1,B2. These are components of normalized full-system eigenvectors at the individual OFF bias. The plot shows compensated minus nominal weight. Negative changes mean reduced A-rail participation. Four-state renormalization is absent.

Nominal state	A1 weight	A2 weight	A1+A2 weight
B0L	6.177e-09	1.369e-06	1.376e-06
B1L	4.774e-09	1.334e-06	1.339e-06

All four rail-state components for both rails and all 24 cases are embedded. Participation changes retain the numerical-basis qualification of the underlying state calculation.

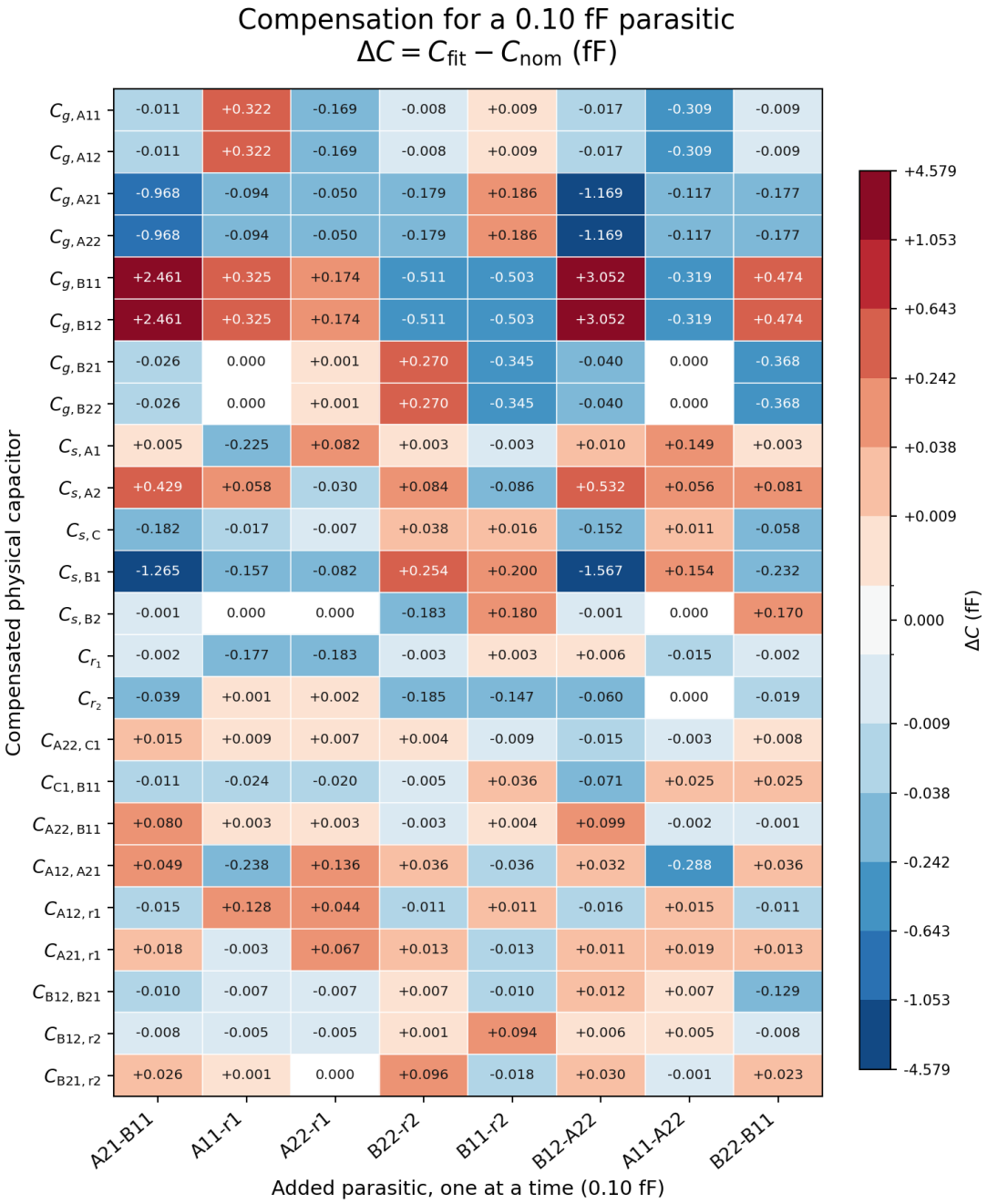
Group 1 compensation | 0.1 fF | %



Percentage boundaries are 0, ±1, ±3, ±5, ±8 and ±10%. End colors cover values beyond ±10%.
Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

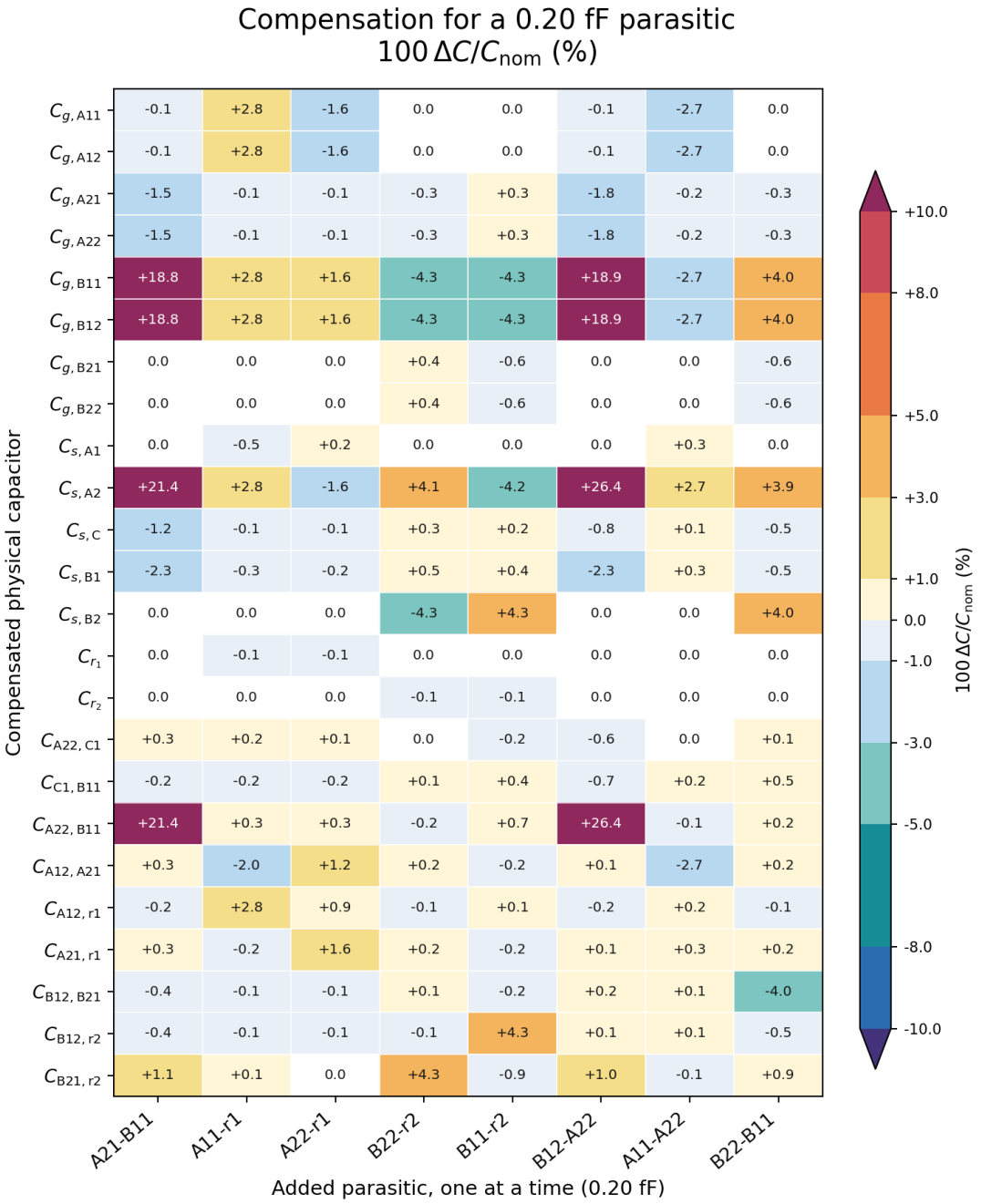
Group 1 compensation | 0.1 fF | fF



The two amplitudes share the same fF color boundaries. Values show signed capacitor adjustments. Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

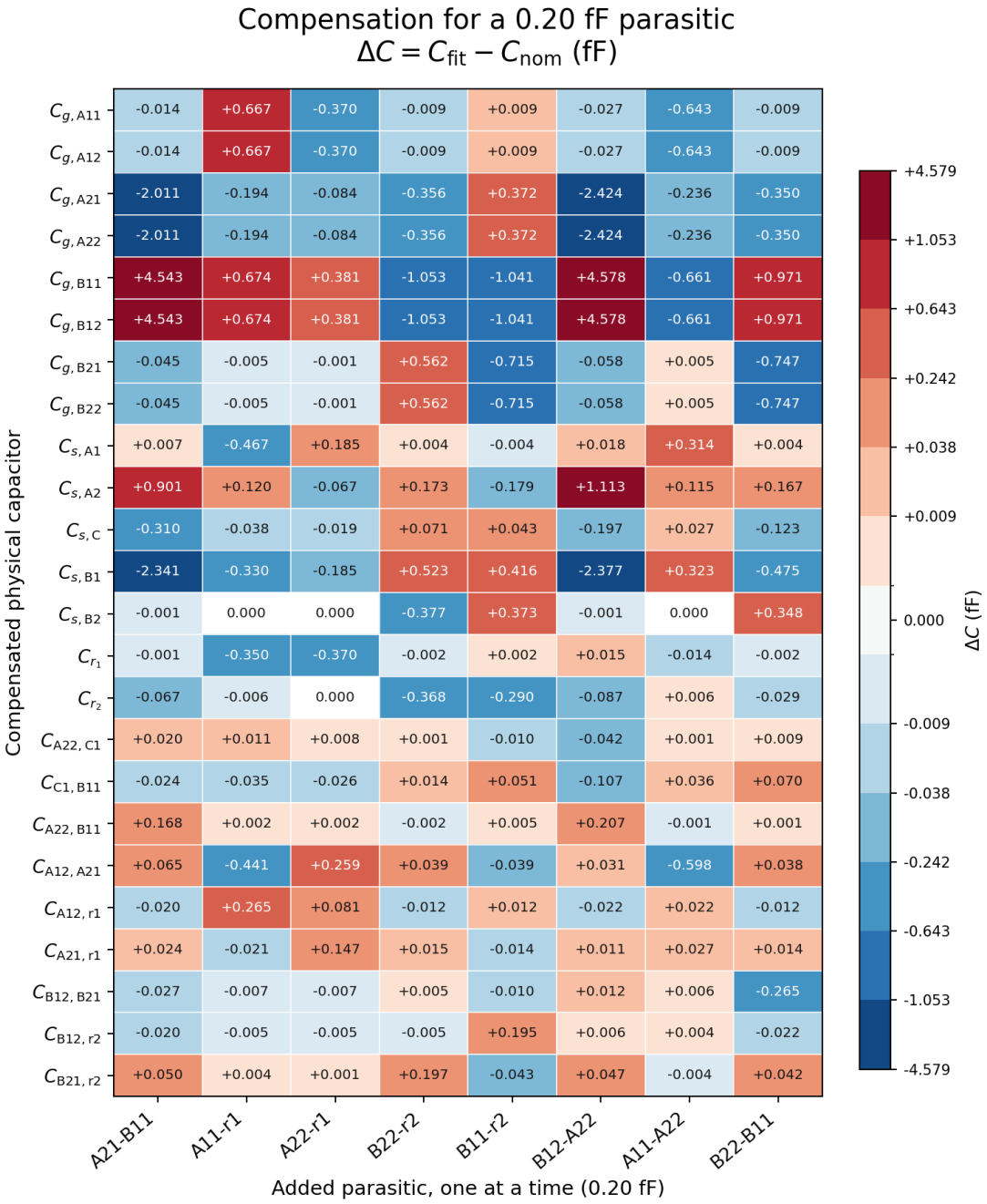
Group 1 compensation | 0.2 fF | %



Percentage boundaries are 0, ±1, ±3, ±5, ±8 and ±10%. End colors cover values beyond ±10%.
Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

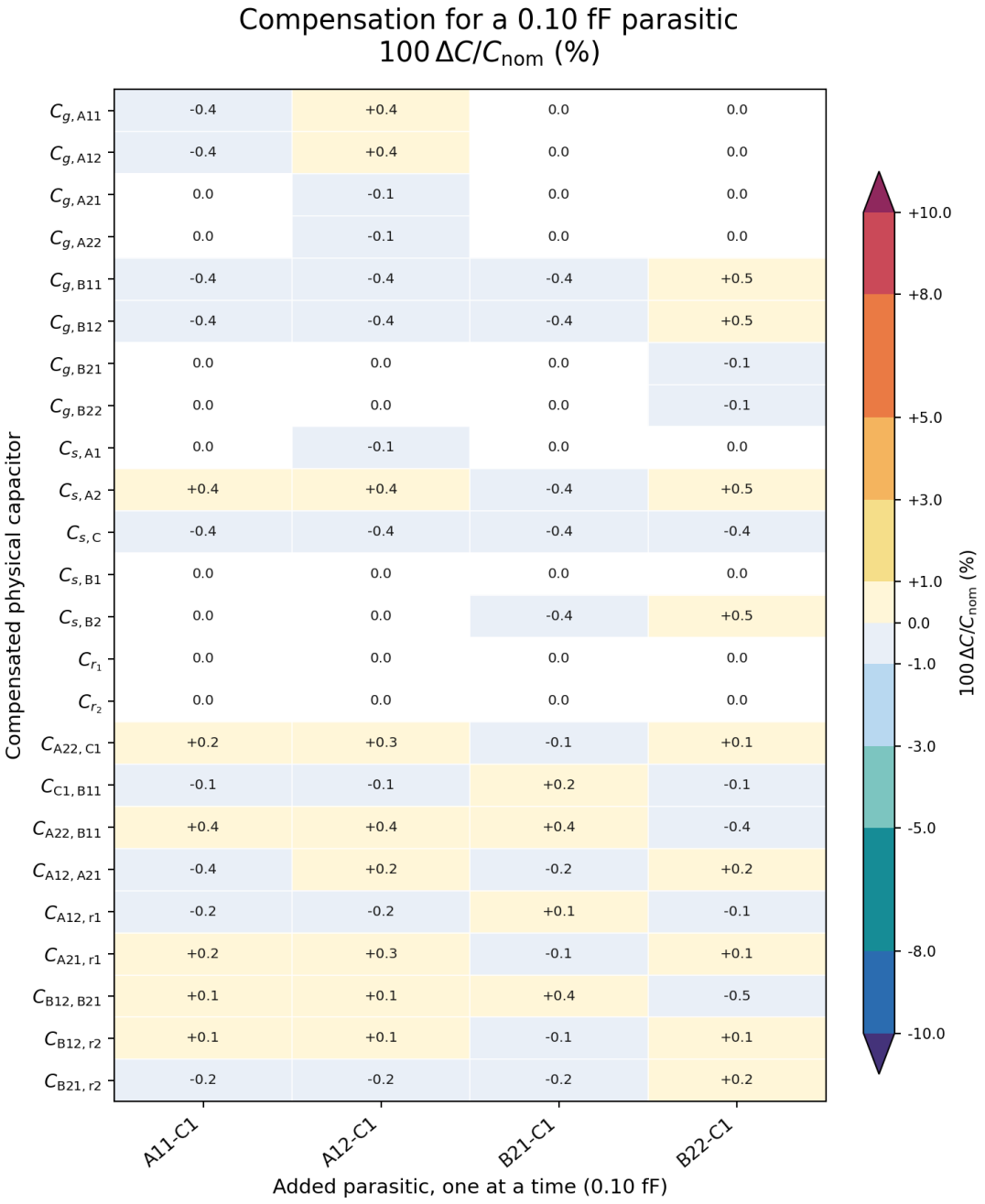
Group 1 compensation | 0.2 fF | fF



The two amplitudes share the same fF color boundaries. Values show signed capacitor adjustments. Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

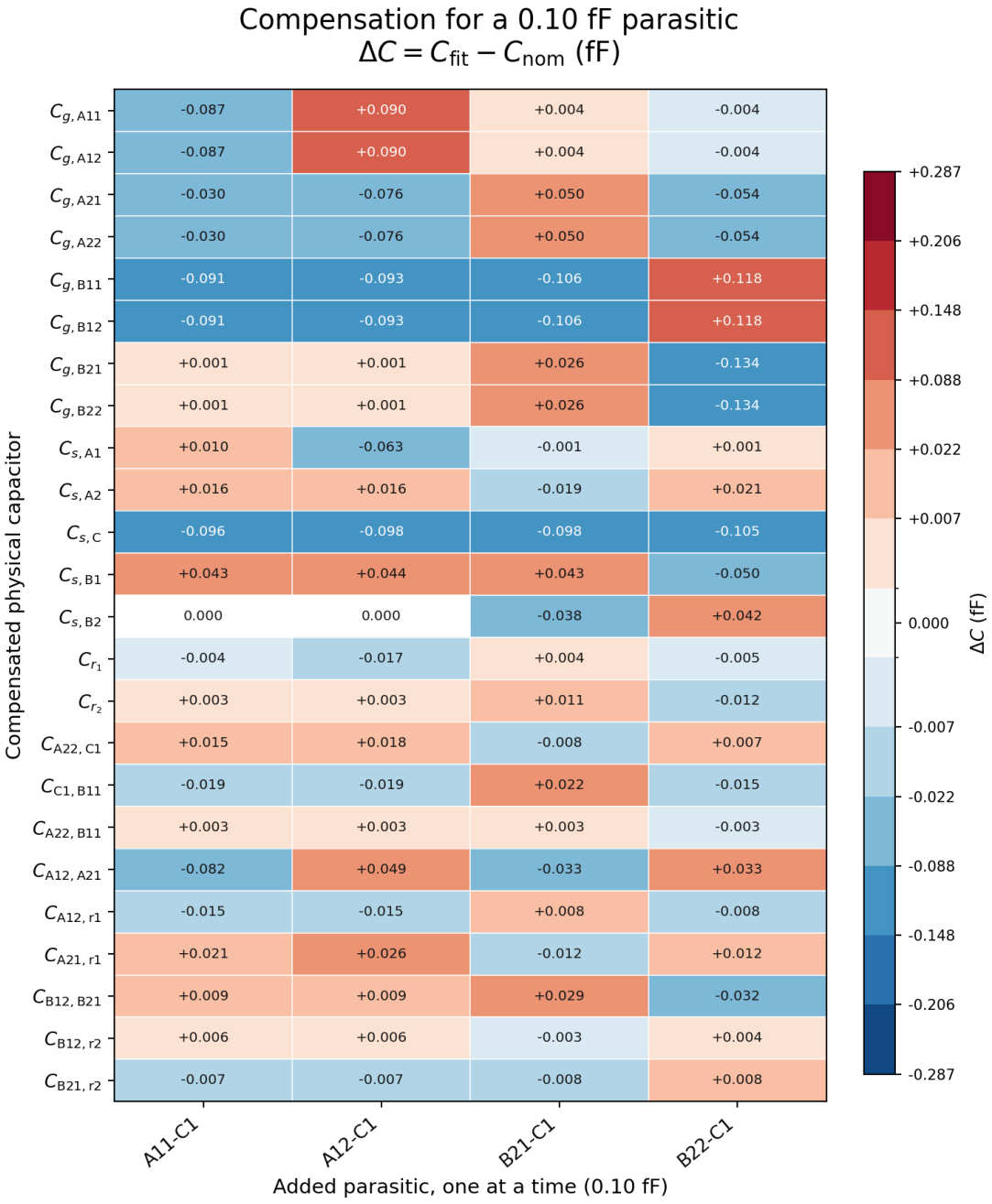
Group 2 compensation | 0.1 fF | %



Percentage boundaries are 0, ±1, ±3, ±5, ±8 and ±10%. End colors cover values beyond ±10%.
Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

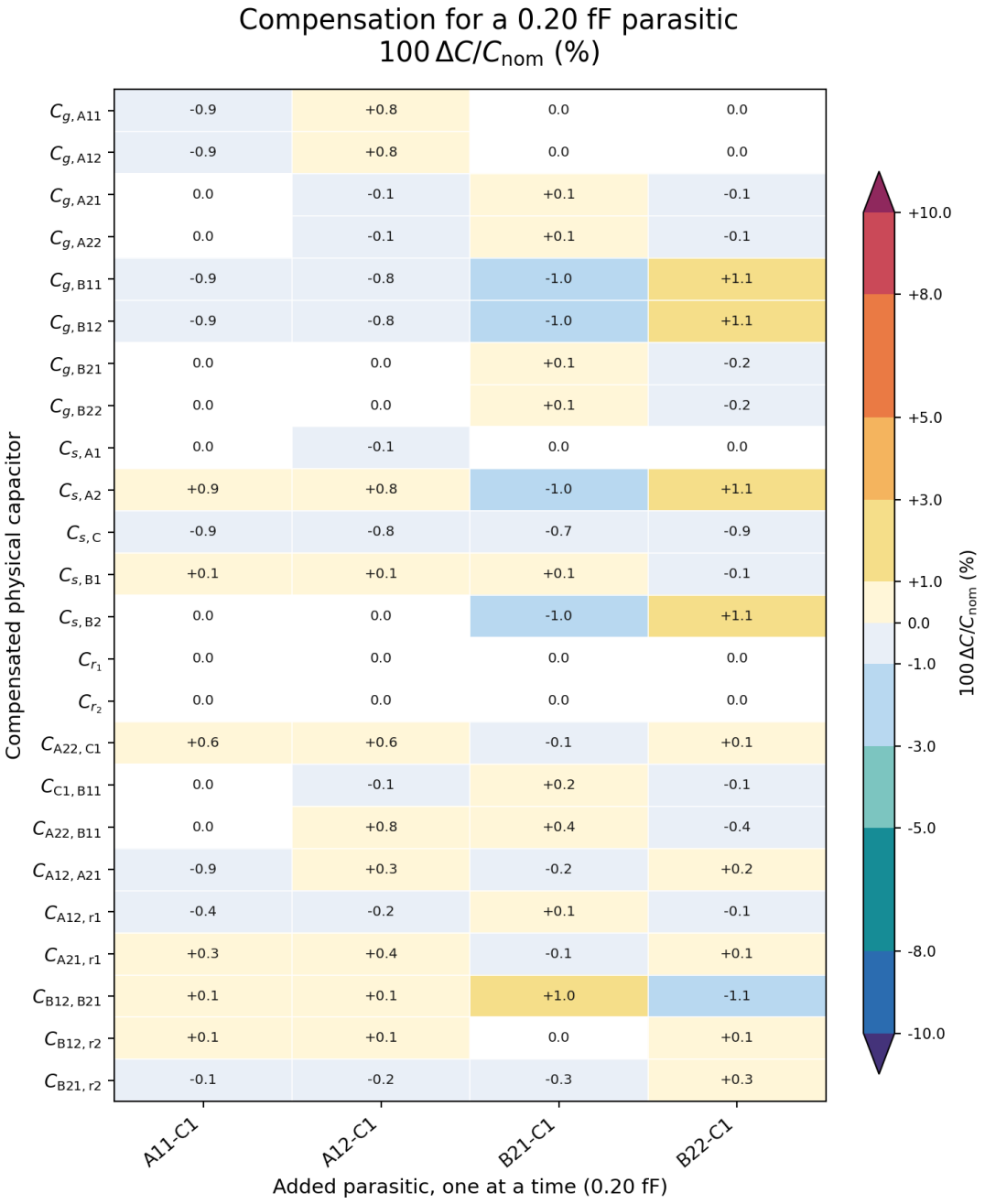
Group 2 compensation | 0.1 fF | fF



The two amplitudes share the same fF color boundaries. Values show signed capacitor adjustments. Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

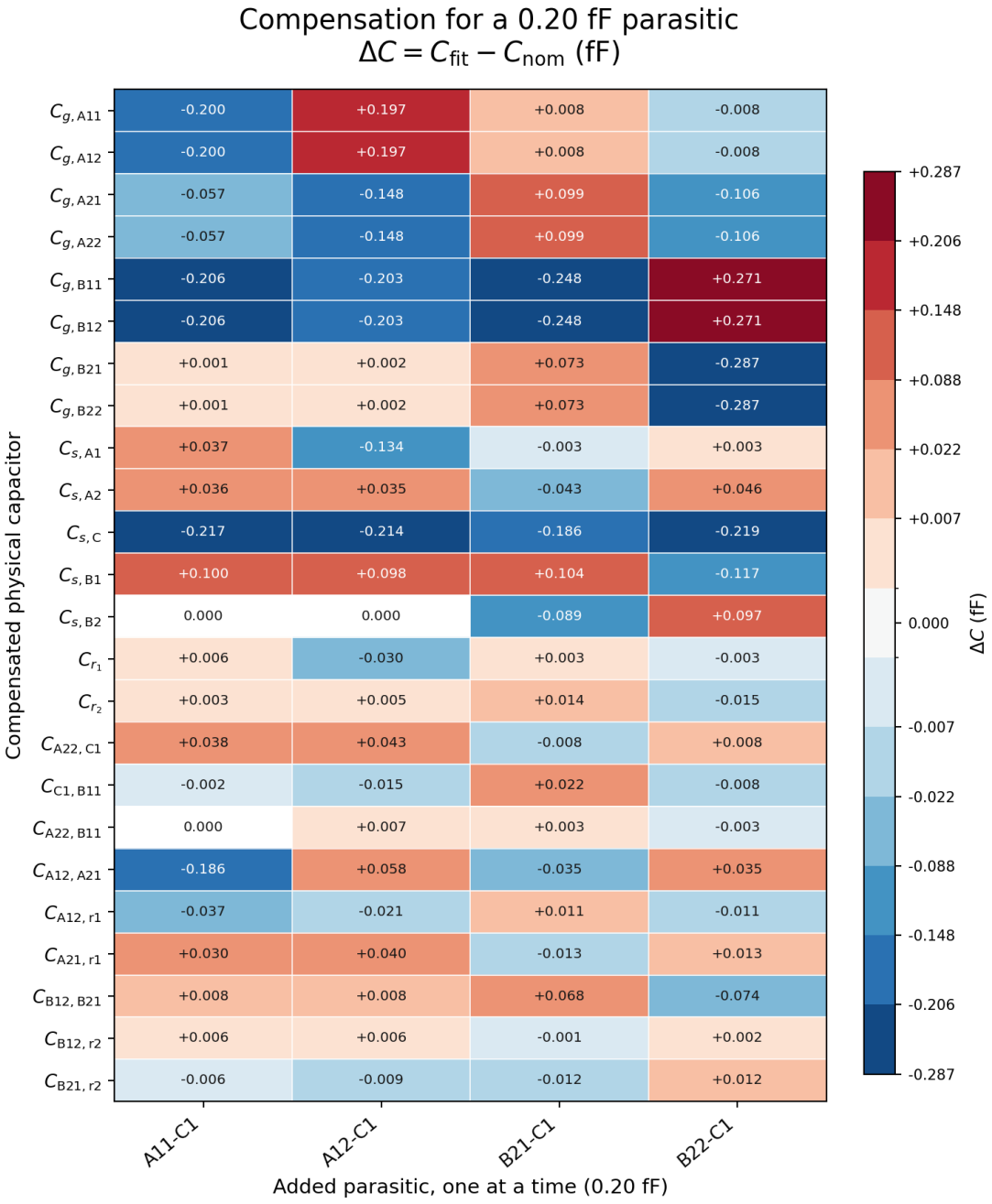
Group 2 compensation | 0.2 fF | %



Percentage boundaries are 0, ±1, ±3, ±5, ±8 and ±10%. End colors cover values beyond ±10%.
Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

Group 2 compensation | 0.2 fF | fF



The two amplitudes share the same fF color boundaries. Values show signed capacitor adjustments. Warm colors show increases. Cool colors show decreases. White marks changes rounding to zero.

Columns are independent single-parasitic cases. Rows are the compensating physical capacitors of the same geometries used for the quantum results. Percentages use one decimal and fF values use three. Percentage bands are 0, +/-1, +/-3, +/-5, +/-8 and +/-10%, with end colors covering larger changes. The two amplitudes share each unit's color scale within the same group.

Numerical accuracy and execution time

Check	Worst measured change or result
Raw gate fidelity basis change	3.395e-05 probability
Selected gate score basis change	2.765e-07 probability
Mean survival basis change	3.368e-05 probability
Physical OFF coupling at identical bias	32.52 Hz between production and audit
Logical exchange at identical bias	10.74 Hz between production and audit
Secular ZZ at identical bias	0.37 Hz between production and audit

Production basis is local6/pair12/C5 with charge cutoff ncut35. Audit basis is local7/pair15/C6 with ncut40. Each resonator retains states 0,1,2. All basis comparisons hold the reported controls fixed. The gate probability target is 1e-4, or 0.01 percentage point. Within-basis OFF acceptance is |physical g| below 1 Hz. A finite production/audit difference above 1 Hz leaves cross-basis residual certification unresolved. Values remain reported with that qualification.

Execution measure	Minutes
Campaign elapsed wall time	65.75
Workers	2
gate and fixed idle	30.42
off search	70.01
logical residual	14.22
off basis check	2.30
calibrated gate basis	14.23
Fixed-bias logical postchecks, summed	2.35

Stage times sum worker execution time. Parallel worker durations overlap and exceed elapsed wall time. The frozen candidate geometry and case identity are checked before reporting. Nominal inputs, minimum-change fits, all quantum case records, basis checks and unrounded controls are embedded for reproduction. Local convergence of the capacitor fit carries the qualification stated on page 2.